User Scenarios List

short description of all the user scenarios

Nr	Name	Description
US_01	Create account	User: A new user registers on the Librario platform by filling out a registration form with name, email, and password.
US_02	Log in	User: All users (Member, Staff, Admin) log in using email and password.
US_03	Log out	User: A logged-in user logs out to end their session securely.
US_04	Update profile	User: A logged-in user updates their display name or annual reading goal.
US_05	Change password	User: A logged-in user changes their password by verifying their current one first.
US_06	Reset password	User: A user who forgot their password resets it via an email link.
US_07	Upload profile picture	User: A logged-in user uploads a photo to personalize their profile.
US_08	View profile	User: A logged-in user views their own account details (name, email, role, goal, join date).
US_09	Manage user accounts	Administrator: The admin searches for users, views their details, and changes role or status.
US_10	Deactivate account	Administrator: The admin deactivates a user account so the user can no longer log in.

User Scenarios Extended

US_01: Create account:

1. User navigates to the registration page.
2. User enters name, email, and password.
3. System validates all fields and checks email uniqueness.
4. System hashes the password and creates the account.
5. User is logged in and redirected to the dashboard.
6. If email already exists: System displays error 'Email already exists'.
7. If required fields are empty: Form highlights missing fields.

US_02: Log in:

1. User navigates to the login page.
2. User enters email and password.
3. System verifies credentials using BCrypt.
4. System stores session data and redirects to the dashboard.
5. If credentials are invalid: System shows 'Invalid email or password'.
6. If account is deactivated: System shows 'Account deactivated. Contact an administrator.

US_03: Log out:

1. User clicks the Logout button in the navigation bar.
2. System clears all session data from sessionStorage.
3. User is redirected to the login page.
4. If session already expired: User is silently redirected to login.

US_04: Update profile:

1. User navigates to the Stats page.
2. User edits the name or reading goal field.
3. User submits the form.
4. System saves the changes and shows a confirmation message.
5. If name field is left blank: Existing name is preserved.

US_05: Change password:

1. User navigates to the Stats page, Change Password section.
2. User enters current password, new password, and confirmation.
3. System verifies the current password.
4. System saves the new BCrypt-hashed password.
5. Success message is displayed.
6. If current password is wrong: Error 'Current password is incorrect'.
7. If new passwords do not match: Inline validation error shown.
8. If new password is under 6 characters: Error message displayed.

US_06: Reset password:

1. User clicks 'Forgot password' on the login page.
2. User enters their registered email address.
3. System sends a time-limited reset link to that email.
4. User clicks the link and enters a new password.
5. System saves the new hashed password.
6. If email not found: Generic message shown (prevents enumeration).
7. If link expired: User must request a new reset link.

US_07: Upload profile picture:

1. User navigates to profile settings.
2. User selects an image file (JPG or PNG).
3. System validates the file format and size (max 2 MB).
4. System resizes and stores the image.
5. Profile picture is updated and displayed.
6. If file is too large: Error message with size limit shown.
7. If format is unsupported: Format error message displayed.

US_08: View profile:

1. User navigates to their profile page.
2. System retrieves the user's name, email, role, reading goal, and join date.
3. Data is displayed in a structured layout.
4. If data is temporarily unavailable: A friendly error message is shown.

US_09: Manage user accounts:

1. Admin navigates to the Admin Panel.
2. Admin searches for users by name or email.
3. System returns a list of matching users with details.
4. Admin can change a user's role or toggle their active/inactive status.
5. Changes are saved immediately.
6. If no users match the search: 'No users found' message is shown.
7. If a non-admin accesses the panel: Redirected to the dashboard.

US_10: Deactivate account:

1. Admin finds the target user in the Admin Panel.
2. Admin clicks the 'Deactivate' button.
3. System sets IsActive = false for that user.
4. User can no longer log in.
5. Admin can reactivate the account by clicking 'Activate'.
6. If admin attempts to deactivate their own account: System prevents the action.

Use Cases

UC_01: Create account

UC Name	UC_01 — Create account
Dependency	None
Actors	Primary actor: User (new visitor) Secondary actor: Email Validation Service
Preconditions	User has not previously registered. System is accessible.
Description of the Main Sequence	1. User opens the registration page. 2. User enters name, email, and password. 3. System validates all fields for completeness and email uniqueness. 4. System hashes the password with BCrypt and creates the user record. 5. User is logged in and redirected to the dashboard.
Description of the Alternative Sequence	• Email already registered: System returns error 'Email already exists'. • Required fields missing: Form highlights empty fields. • Password too short: Inline hint shown.
Non-functional requirements	Registration must complete within 3 seconds. Passwords must be stored as BCrypt hashes. System must prevent duplicate email addresses.
Postconditions	A new Member account is created and the user is authenticated.

UC_02: Log in

UC Name	UC_02 — Log in
Dependency	UC_01 — account must exist
Actors	Primary actor: User Secondary actor: None
Preconditions	User has a registered, active account.

Description of the Main Sequence	1. User navigates to the login page. 2. User enters email and password. 3. System verifies credentials using BCrypt. 4. System stores session data (userId, name, role) and redirects to dashboard.
Description of the Alternative Sequence	• Invalid credentials: Error 'Invalid email or password'. • Account deactivated: Error 'Account deactivated. Contact an administrator'.
Non functional requirements	Login response must be under 2 seconds. Credentials must never be stored in plain text.
Postconditions	User is authenticated and session is active.

UC_03: Log out

UC Name	UC_03 — Log out
Dependency	UC_02
Actors	Primary actor: User Secondary actor: None
Preconditions	User is currently logged in.
Description of the Main Sequence	1. User clicks the Logout button in the header. 2. System clears session storage. 3. User is redirected to the login page.
Description of the Alternative Sequence	• Session already expired: User is silently redirected to login.
Non functional requirements	Logout must complete instantly (< 0.5s). All session data must be cleared completely.
Postconditions	Session is terminated. User must log in again to access the platform.

UC_04: Update profile

UC Name	UC_04 — Update profile
Dependency	UC_02
Actors	Primary actor: User Secondary actor: None
Preconditions	User is logged in.
Description of the Main Sequence	1. User navigates to Stats page. 2. User edits name and/or reading goal fields. 3. User submits the changes. 4. System validates and saves. Confirmation message shown.
Description of the Alternative Sequence	• Name field submitted empty: Existing name is preserved.
Non functional requirements	Changes must be saved within 2 seconds.

Postconditions	Profile is updated with the new values.

UC_05: Change password

UC Name	UC_05 — Change password
Dependency	UC_02
Actors	Primary actor: User Secondary actor: None
Preconditions	User is logged in and knows their current password.
Description of the Main Sequence	1. User navigates to the Change Password section on Stats page. 2. User enters current password, new password, and confirmation. 3. System verifies current password with BCrypt. 4. System saves new BCrypt hash. 5. Success message is displayed.
Description of the Alternative Sequence	• Current password wrong: Error 'Current password is incorrect'. • New passwords do not match: Inline validation error. • New password under 6 characters: Error message shown.
Non functional requirements	New password must be stored as BCrypt hash. Change must complete within 2 seconds.
Postconditions	Password is updated. Next login requires the new password.

UC_06: Reset password

UC Name	UC_06 — Reset password
Dependency	None — user is not logged in
Actors	Primary actor: User Secondary actor: Email Service
Preconditions	User has a registered email address.
Description of the Main Sequence	1. User clicks 'Forgot password' on login page. 2. User enters their email address. 3. System sends a time-limited reset link to that email. 4. User clicks the link and enters a new password. 5. System saves the new BCrypt hash.
Description of the Alternative Sequence	• Email not found: Generic message shown (prevents user enumeration). • Reset link expired: User must request a new link.
Non functional requirements	Reset link must expire after 30 minutes and be invalidated after first use.
Postconditions	Password is updated. Old sessions are invalidated.

UC_07: Upload profile picture

UC Name	UC_07 — Upload profile picture

Dependency	UC_02
Actors	Primary actor: User Secondary actor: File Storage Service
Preconditions	User is logged in.
Description of the Main Sequence	1. User navigates to profile settings. 2. User selects an image file (JPG/PNG). 3. System validates format and size (max 2 MB). 4. System resizes the image to 200×200px and stores it. 5. Profile picture is updated and displayed.
Description of the Alternative Sequence	• File too large: Error message with size limit shown. • Unsupported format: Format error message displayed.
Non functional requirements	Upload must complete within 5 seconds. Image must be resized server-side.
Postconditions	Profile picture is saved and displayed on the user's profile.

UC_08: View profile

UC Name	UC_08 — View profile
Dependency	UC_02
Actors	Primary actor: User Secondary actor: None
Preconditions	User is logged in.
Description of the Main Sequence	1. User navigates to their profile page. 2. System retrieves name, email, role, reading goal, and join date. 3. Data is displayed in a structured layout.
Description of the Alternative Sequence	• Data temporarily unavailable: Friendly error message shown.
Non functional requirements	Profile page must load within 1.5 seconds.
Postconditions	User can see all their current profile information.

UC_09: Manage user accounts

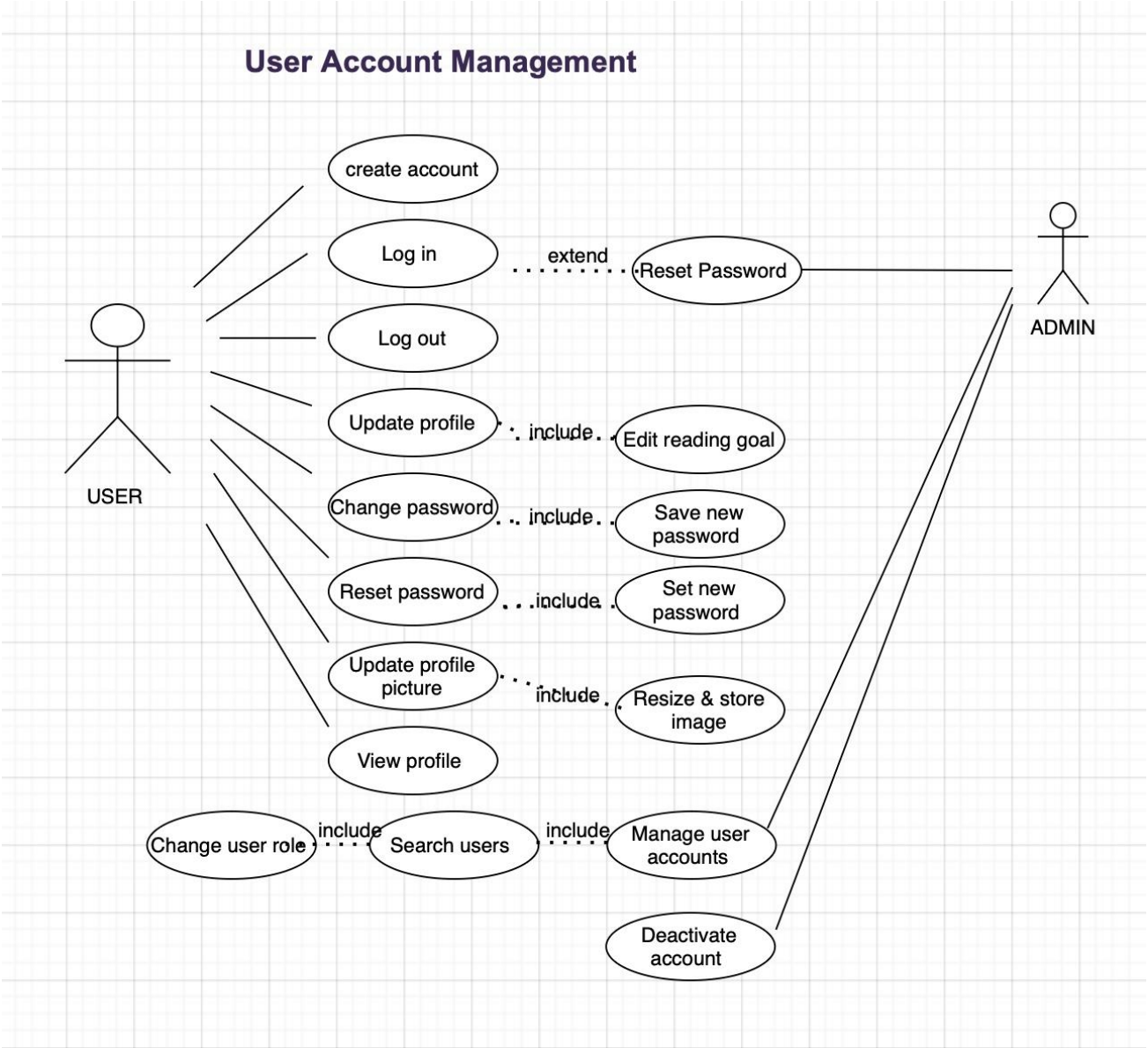
UC Name	UC_09 — Manage user accounts
Dependency	UC_02 — admin must be authenticated
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Admin is logged in. Users exist in the system.
Description of the Main Sequence	1. Admin navigates to Admin Panel. 2. Admin searches for users by name or email. 3. System returns a list of matching users. 4. Admin views user details and can change role or toggle status.

Description of the Alternative Sequence	• No results found: 'No users found' message shown. • Non-admin attempts access: Redirected to dashboard.
Non functional requirements	User list must load within 2 seconds. Role changes must apply immediately.
Postconditions	Target user's role or status is updated in the system.

UC_10: Deactivate account

UC Name	UC_10 — Deactivate account
Dependency	UC_09
Actors	Primary actor: Administrator Secondary actor: None
Preconditions	Admin is logged in. Target user account exists.
Description of the Main Sequence	1. Admin finds the target user in Admin Panel. 2. Admin clicks 'Deactivate'. 3. System sets IsActive = false for that user. 4. User can no longer log in.
Description of the Alternative Sequence	• Admin reactivates account: IsActive reset to true. • Admin attempts to deactivate own account: System prevents the action.
Non functional requirements	Status change must take effect immediately. Action must be logged for audit trail.
Postconditions	Target user account is deactivated. Active sessions are terminated.

Use Case Diagrams



Functional Requirements

Req#	Requirement	Comments	Priority	Date	Reviewed / Approved
FR_01	The system shall allow new users to register with a unique email, name, and password.	Email uniqueness enforced at database level.	1	01/04/2026	Kiara Karriqi
FR_02	The system shall authenticate users using email and BCrypt-hashed password verification.	Plain-text passwords must never be stored or logged.	1	01/04/2026	Kiara Karriqi
FR_03	The system shall store user session data in sessionStorage and clear it completely on logout.	Session includes userId, name, and role.	1	01/04/2026	Kiara Karriqi

FR_04	The system shall allow logged-in users to update their display name and annual reading goal.	Empty name field must preserve existing name.	2	01/04/2026	Kiara Karriqi
FR_05	The system shall require users to verify their current password before setting a new one.	New password must be at least 6 characters.	1	01/04/2026	Kiara Karriqi
FR_06	The system shall send a time-limited password reset link to the user's registered email.	Link must expire after 30 minutes and be invalidated after use.	2	01/04/2026	Kiara Karriqi
FR_07	The system shall accept JPG/PNG profile pictures up to 2 MB and resize them to 200×200px.	Unsupported formats or oversized files must be rejected with an error.	3	01/04/2026	Kiara Karriqi
FR_08	The system shall display the user's name, email, role, reading goal, and account creation date on the profile page.	Data must reflect the latest saved values.	2	01/04/2026	Kiara Karriqi
FR_09	The system shall allow administrators to search users by name or email and change their role or active status.	Role changes must take effect immediately without requiring re-login.	1	01/04/2026	Kiara Karriqi
FR_10	The system shall prevent deactivated users from logging in and must not allow an admin to deactivate their own account.	Deactivation sets IsActive = false immediately.	1	01/04/2026	Kiara Karriqi

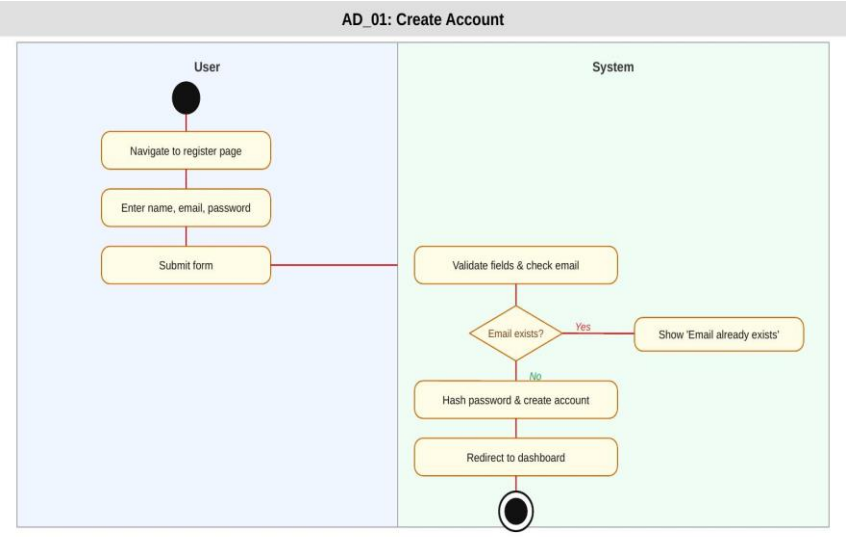
Non-Functional Requirements

Req#	Category	Requirement
NFR_01	Performance	All account operations (login, register, profile update) must respond within 3 seconds under normal load.
NFR_02	Security	All passwords must be stored as BCrypt hashes. Plain-text passwords must never be logged or stored anywhere in the system.
NFR_03	Security	Session data must be cleared completely on logout. Sessions must not persist after browser close.
NFR_04	Security	Password reset links must expire after 30 minutes and be invalidated after first use to prevent reuse attacks.
NFR_05	Usability	Registration and login forms must provide inline validation feedback without a full page reload.
NFR_06	Usability	Error messages must be user-friendly and must not expose internal system details such as stack traces or database errors.

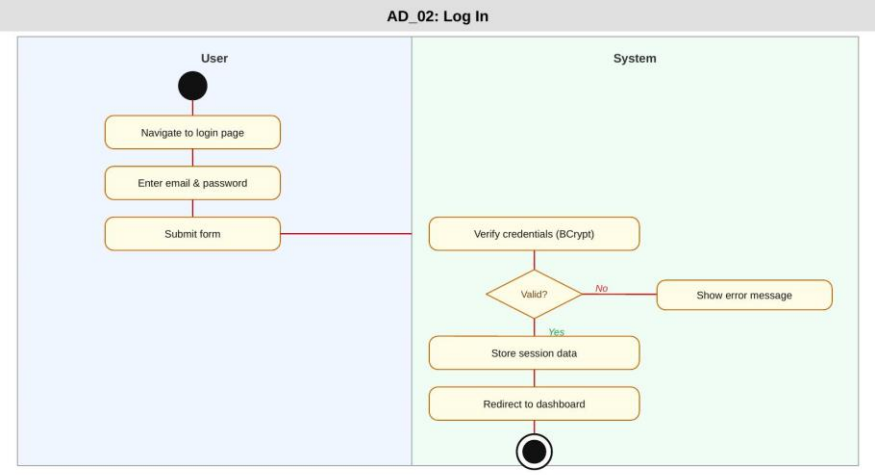
NFR_07	Reliability	The authentication system must maintain 99.5% uptime. Planned maintenance windows must not exceed 1 hour per month.
NFR_08	Scalability	The system must handle at least 100 concurrent authenticated users without performance degradation.
NFR_09	Compliance	User data must be handled in accordance with GDPR. Users must be able to request deletion of their account and personal data.
NFR_10	Auditability	All admin actions (role changes, account deactivations) must be logged with timestamp, admin ID, and affected user ID.

Activity Diagrams

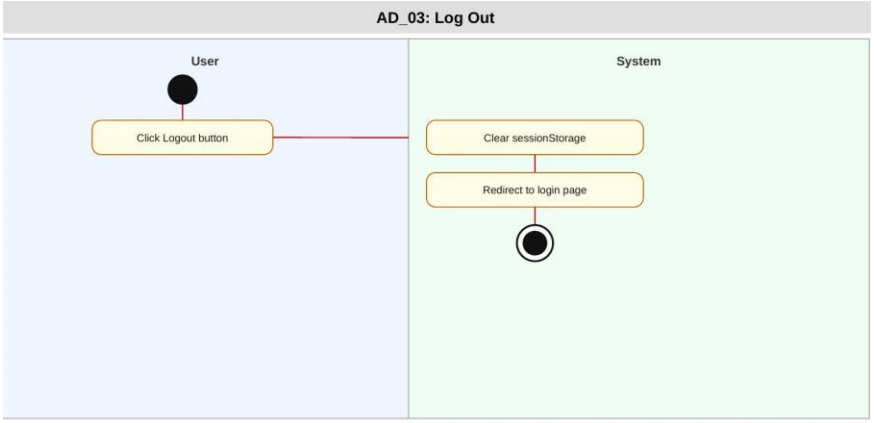
AD_01: Create Account



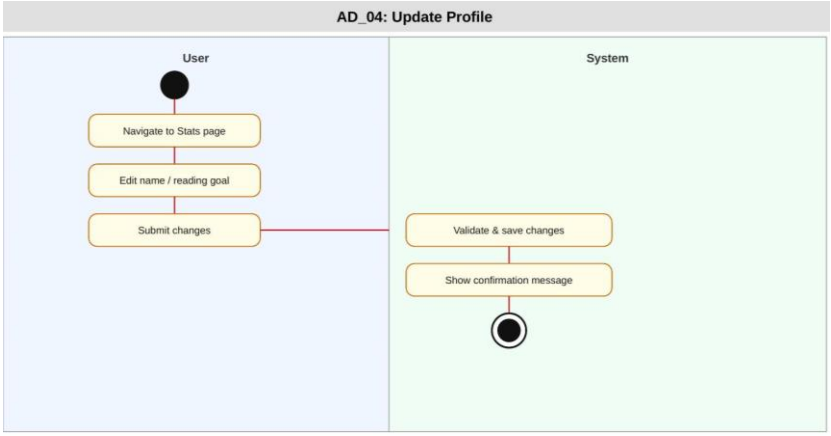
AD_02: Log In



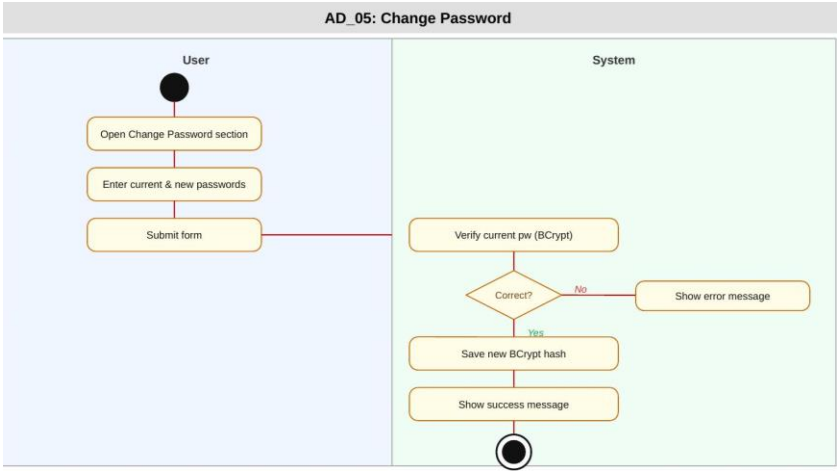
AD_03: Log Out



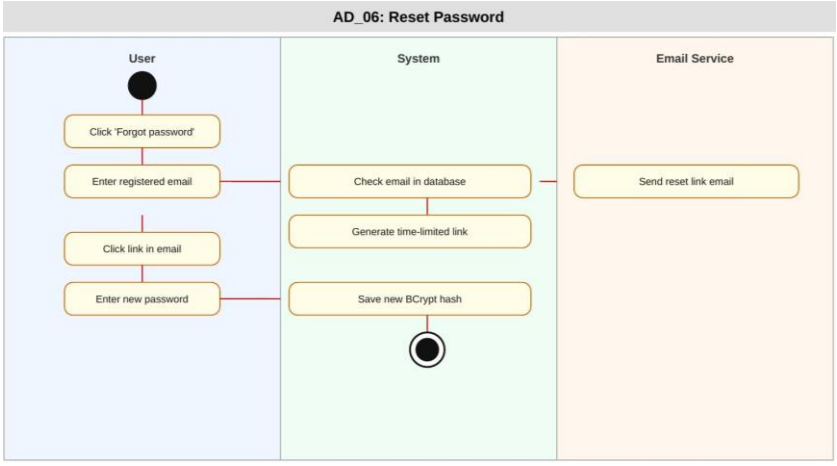
AD_04: Update Profile



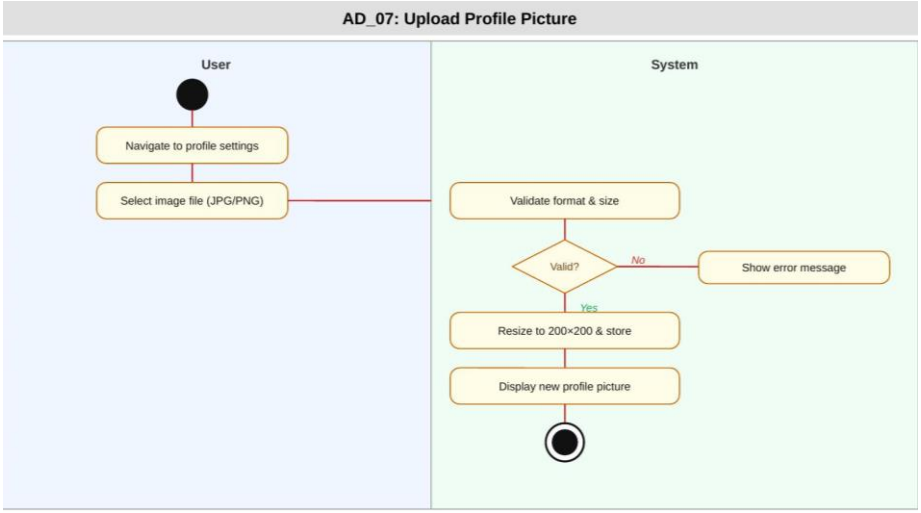
AD_05: Change Password



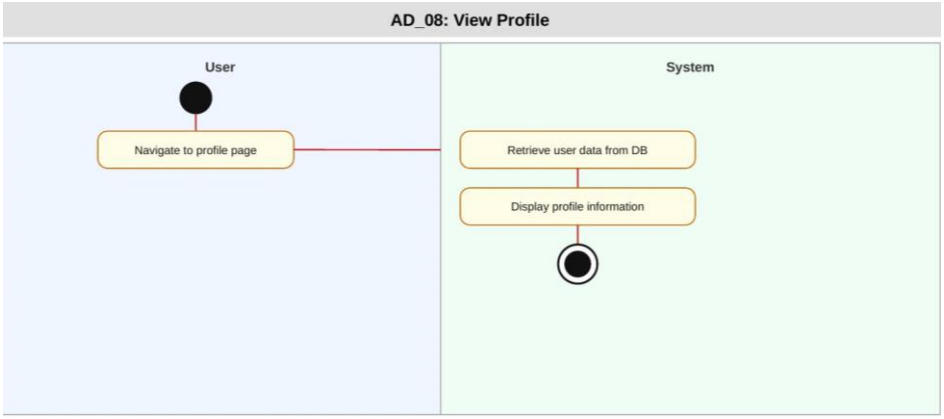
AD_06: Reset Password



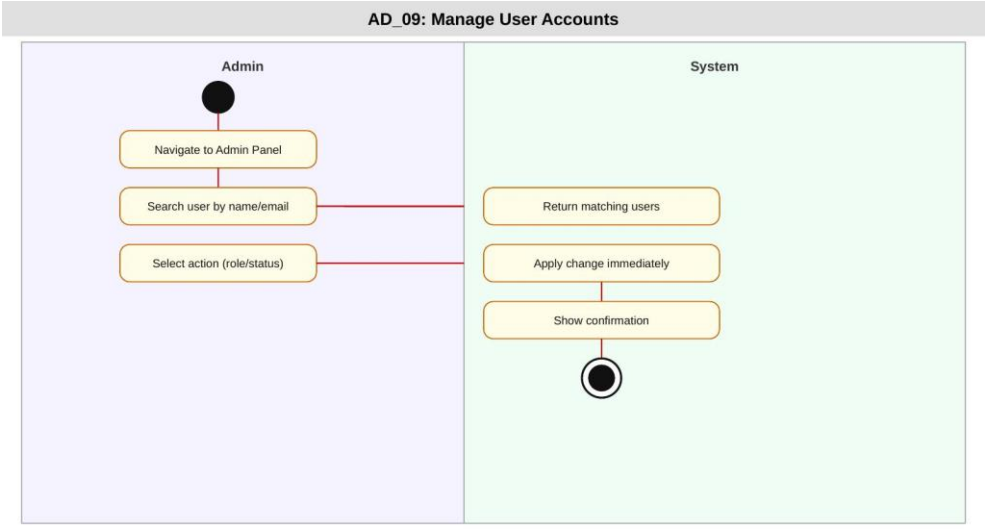
AD_07: Upload Profile Picture



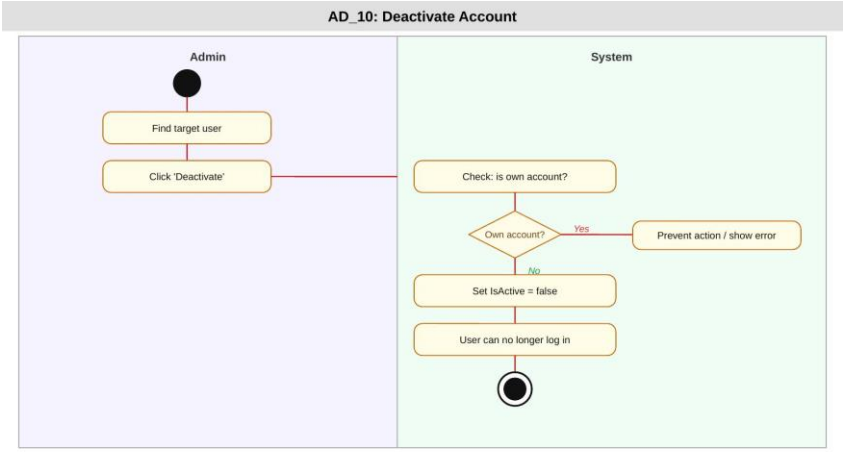
AD_08: View Profile



AD_09: Manage User Accounts

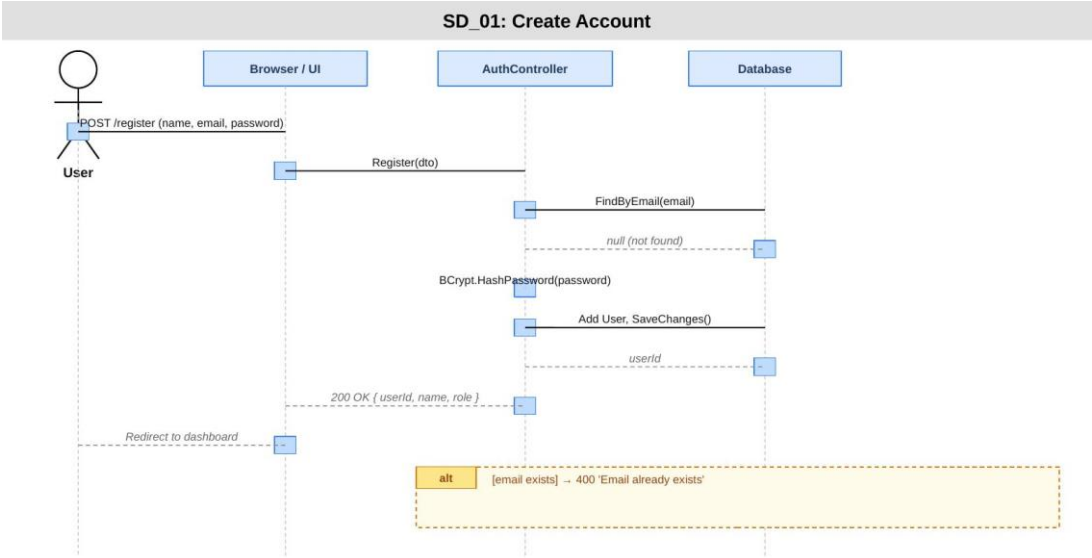


AD_10: Deactivate Account

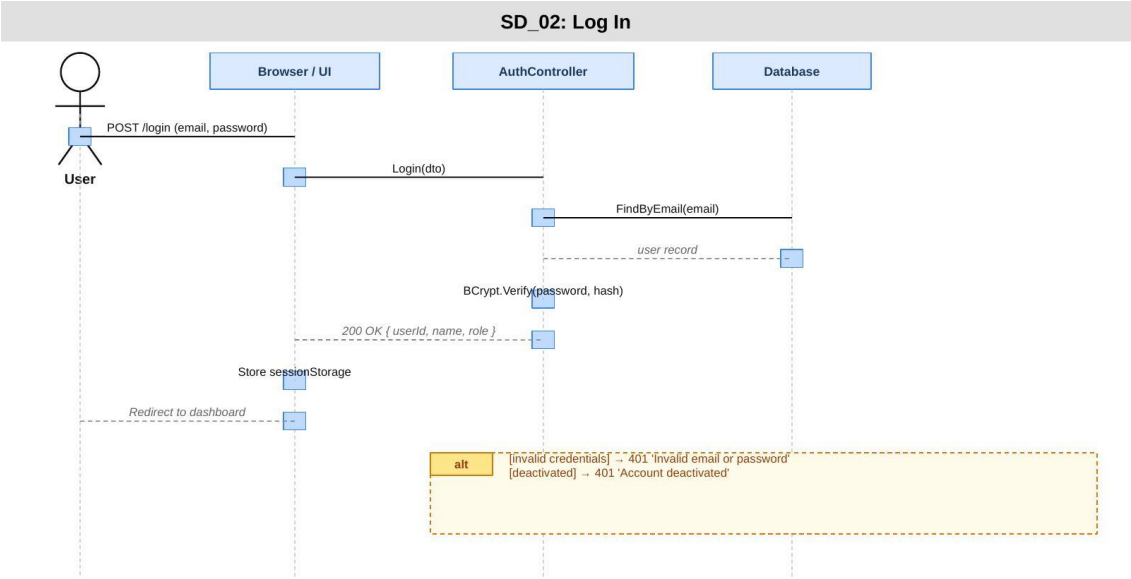


Sequence Diagrams

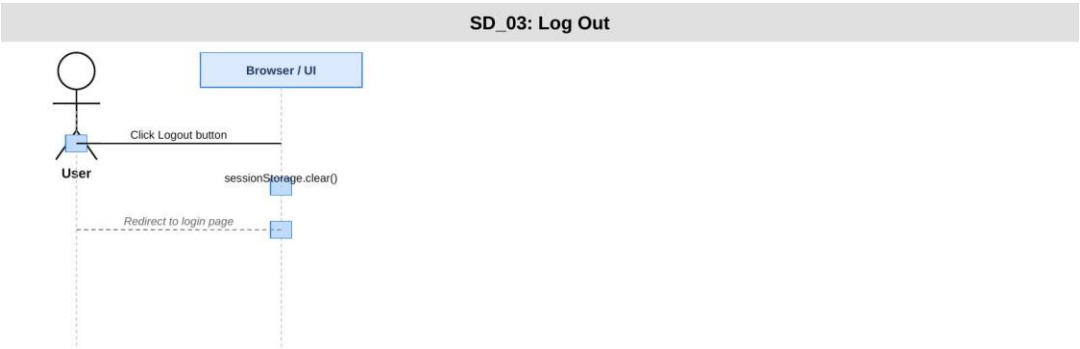
SD_01: Create Account



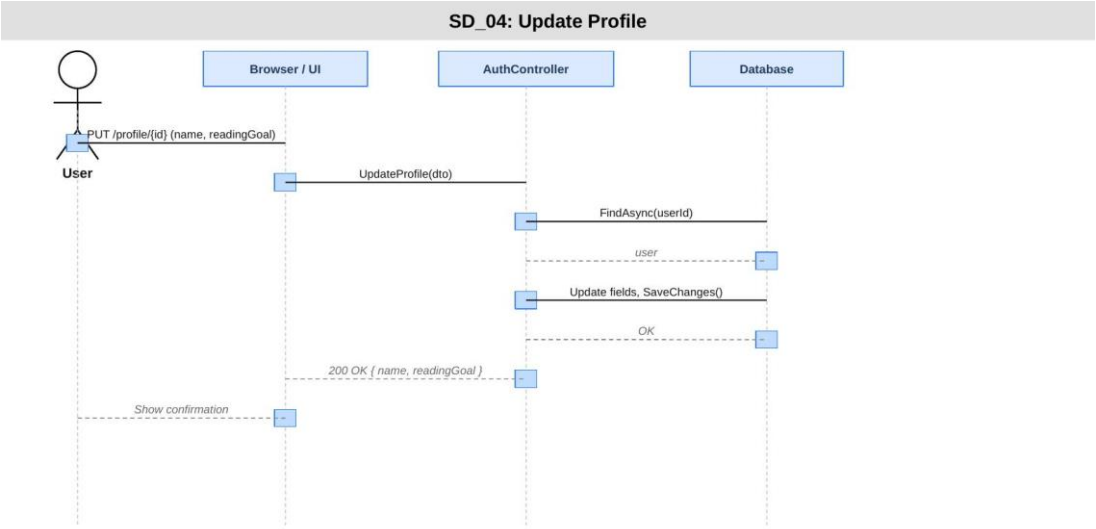
SD_02: Log In



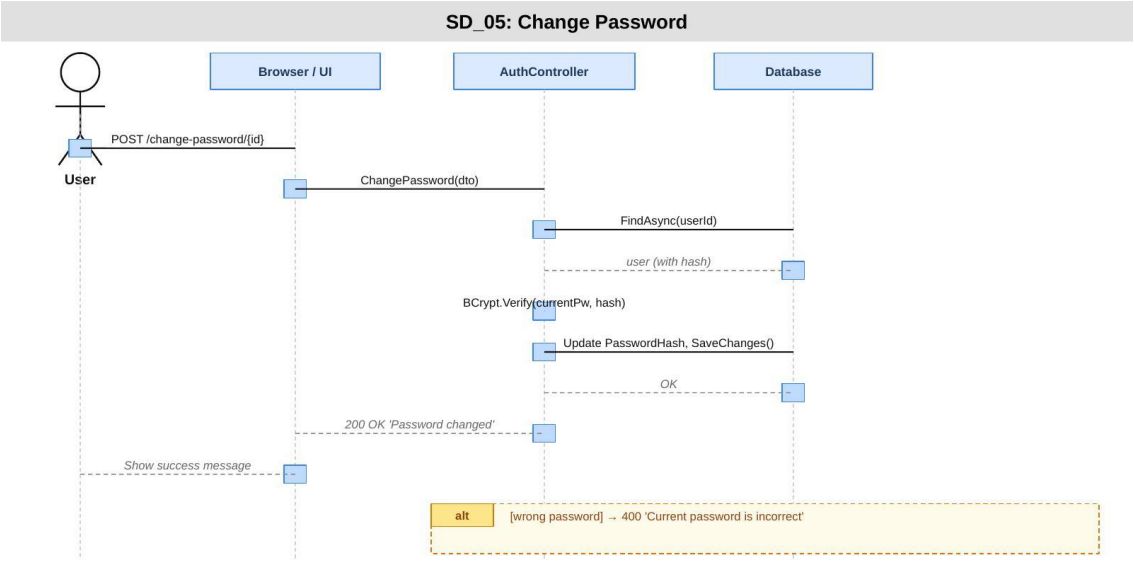
SD_03: Log Out



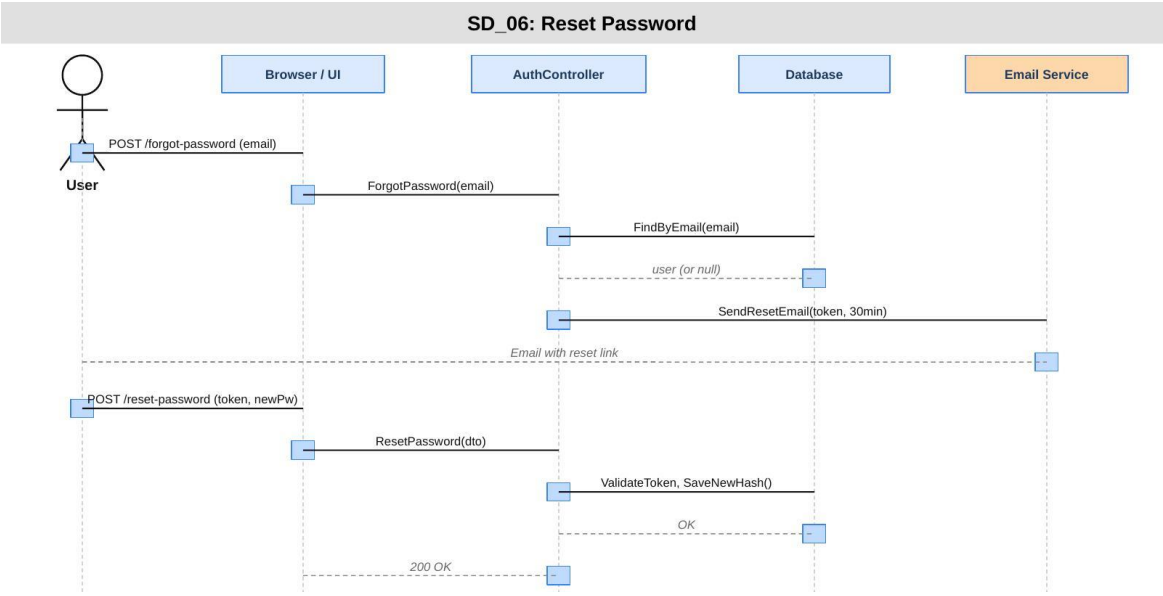
SD_04: Update Profile



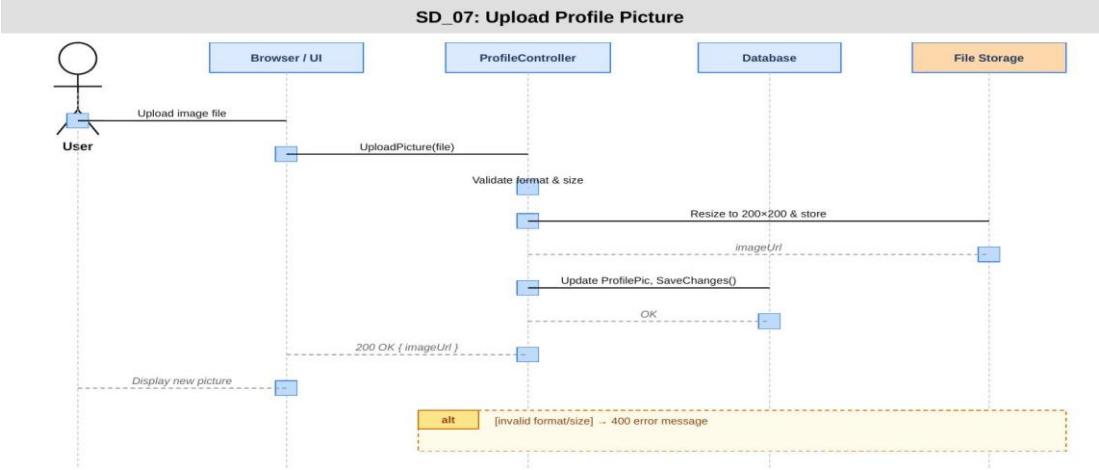
SD_05: Change Password



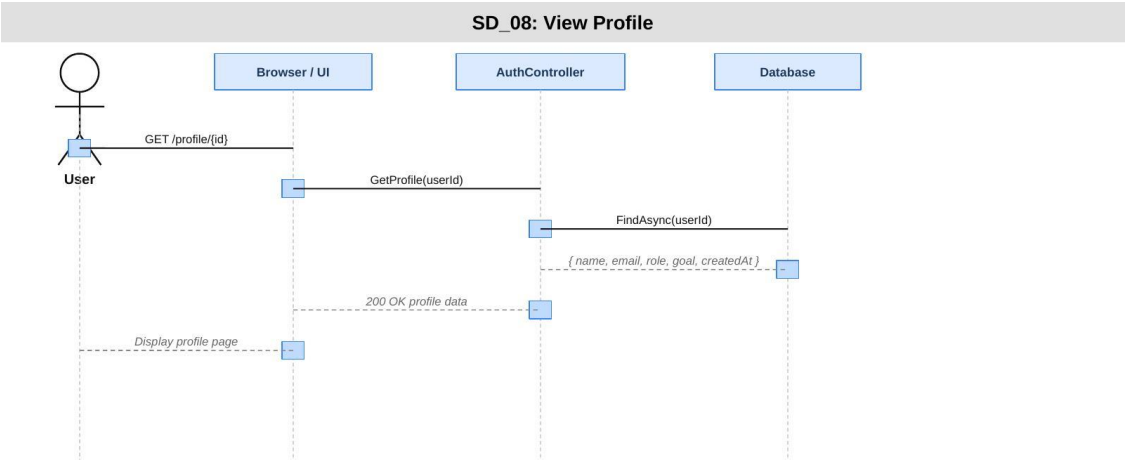
SD_06: Reset Password



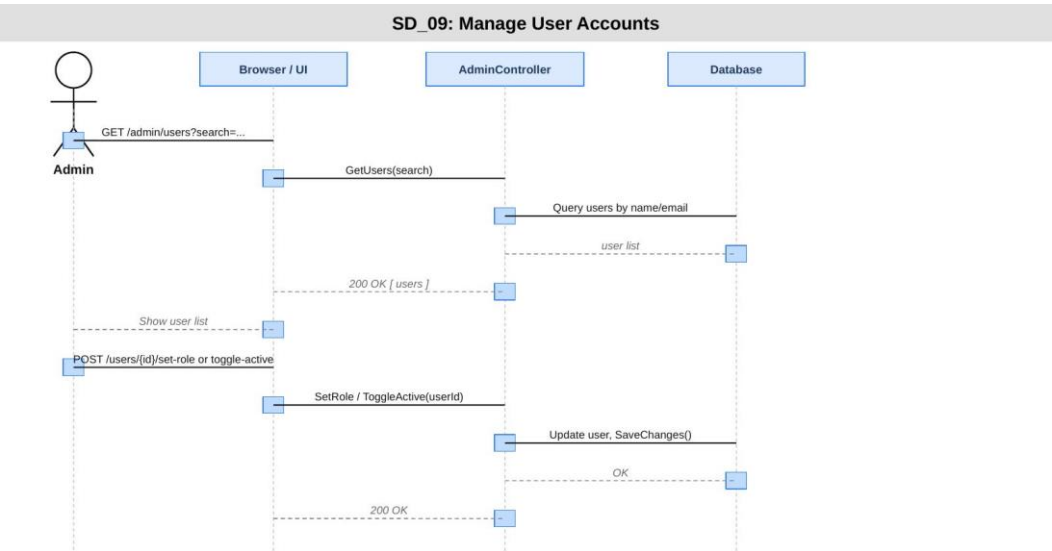
SD_07: Upload Profile Picture



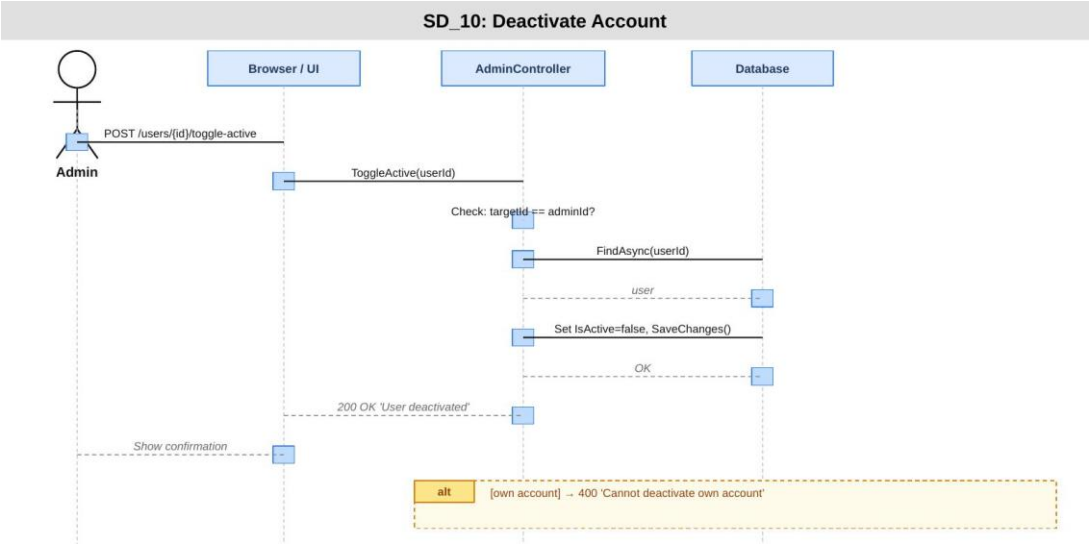
SD_08: View Profile



SD_09: Manage User Accounts

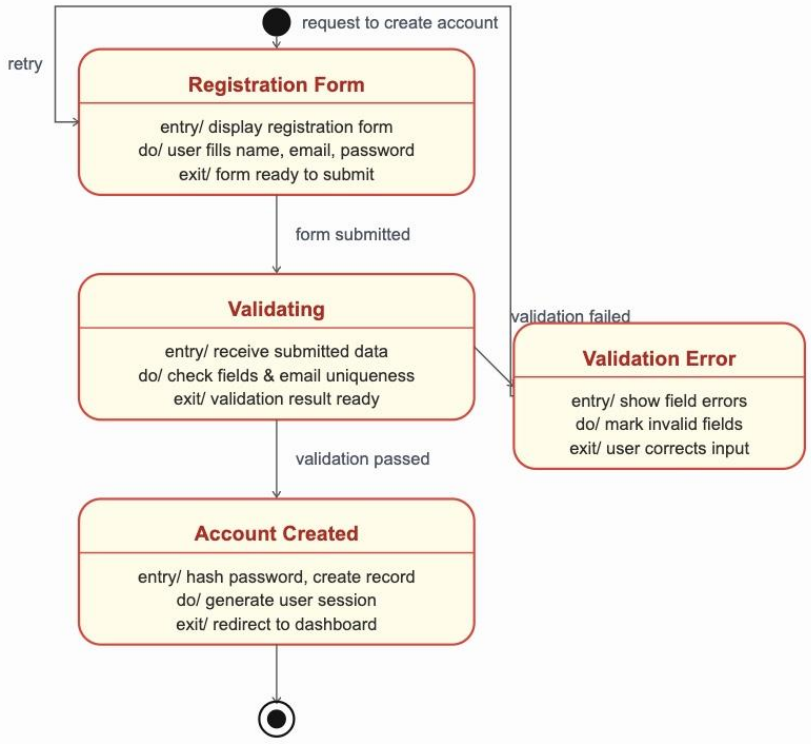


SD_10: Deactivate Account

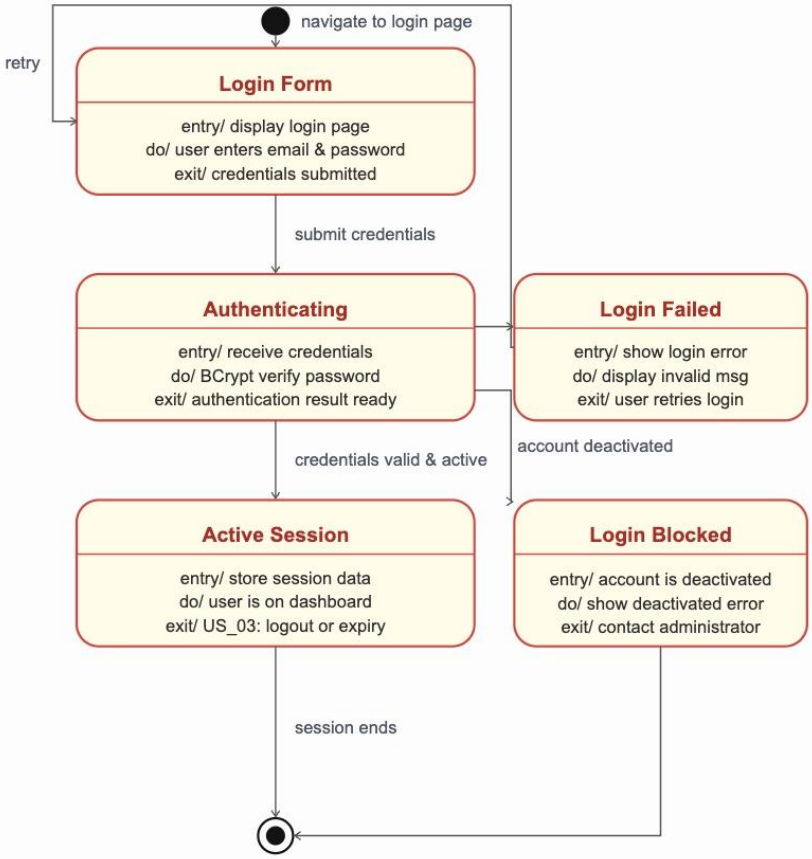


State Diagrams

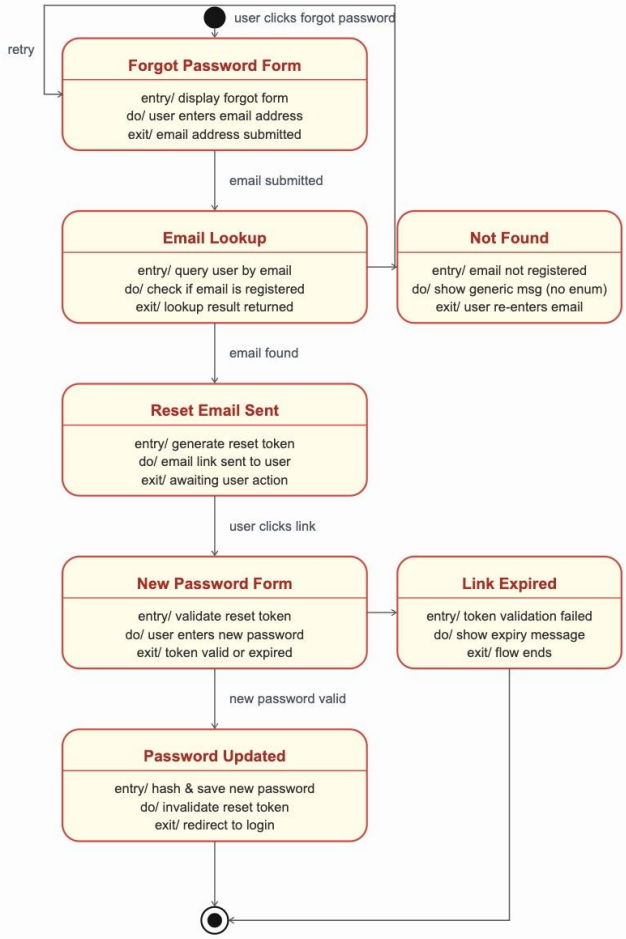
US_01 — Create account



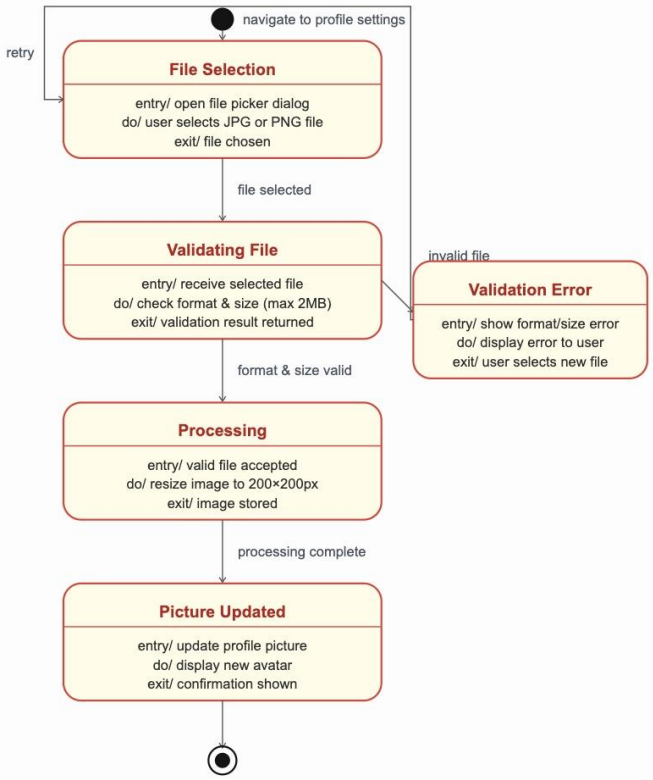
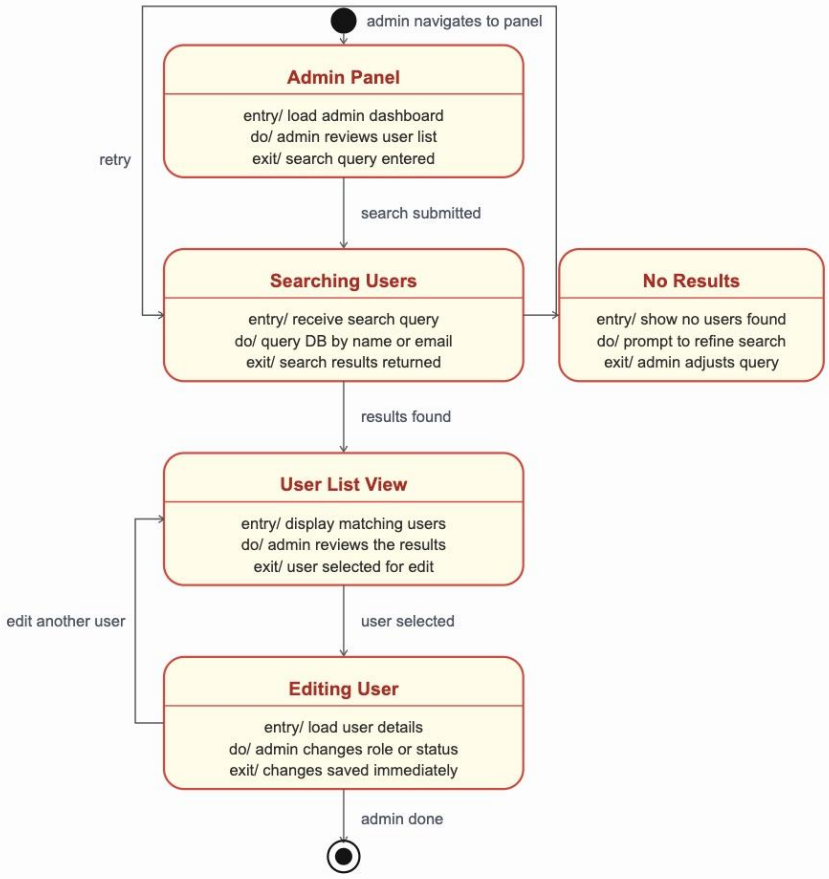
US_02 — Log in



US_06 — Reset password



US_07 — Upload profile picture



US_09 — Manage user accounts